IT1040 – Research Project – 2026

BSc (Hons) in Software Engineering

Deployment Plan - RP

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1. Deployment Overview

This document presents the deployment strategy for the research project **“A Human-Like Agent for Screen-Aware Task Automation.”** The system enables automated interaction with graphical user interfaces by observing a computer screen using an external camera and executing keyboard and mouse actions through a hardware-based Human Interface Device (HID).

The primary objective of the deployment design is to allow the system to operate **externally without installing any software on the target machine**. This approach enables compatibility with legacy systems, secure environments, and black-box testing scenarios where internal system access is restricted.

The deployment architecture combines **hardware components, intelligent processing modules, and external integrations** to enable perception, reasoning, and action execution.

2. System Deployment Architecture

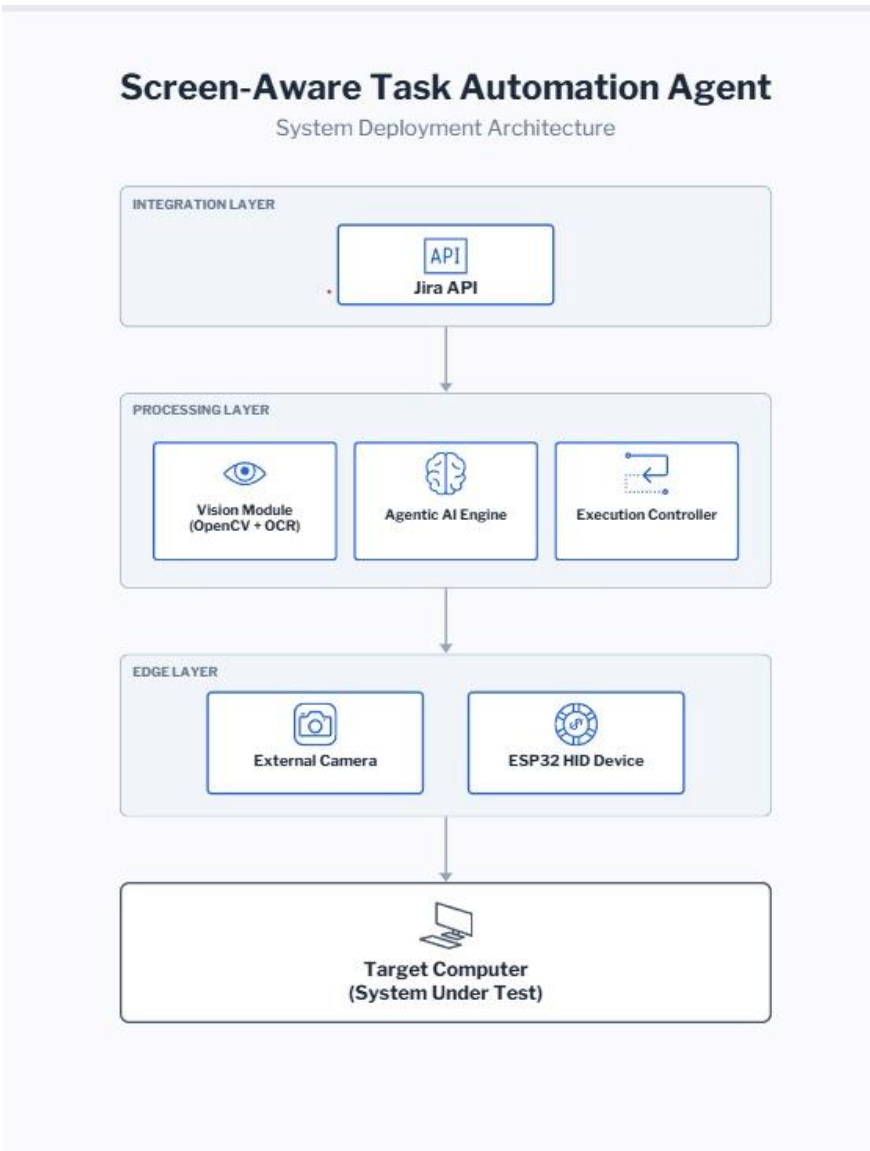
The system is deployed using a **layered hybrid architecture** consisting of three main layers:

- **Edge Layer**
- **Processing Layer**
- **Integration Layer**

Each layer is responsible for a specific part of the automation workflow, from capturing visual information to executing actions and recording results.

3. Deployment Architecture Diagram

The following diagram illustrates the overall system deployment architecture.



This architecture separates the **hardware interaction layer from the intelligent decision-making components**, allowing flexible deployment and scalability.

4. Edge Layer Deployment

The **Edge Layer** consists of hardware components responsible for capturing visual data and performing physical input actions.

- **External Camera**

  An external camera is positioned facing the monitor of the target computer. The camera continuously captures screen frames and provides visual input to the vision processing module.

  This enables the system to **observe the graphical user interface without accessing the operating system directly**.

- **ESP32 HID Controller**

  The ESP32 microcontroller is connected to the target computer through a USB connection and configured to function as a **Human Interface Device (HID)**.

  The device emulates:

  - Keyboard inputs
  - Mouse movements
  - Mouse clicks

  This allows the system to interact with the computer exactly like a human user would.

5. Processing Layer Deployment

The **Processing Layer** contains the intelligent modules responsible for interpreting visual information and determining the next actions.

These modules run on a **dedicated control machine** separate from the target computer.

Vision Processing Module

The vision module processes captured screen images using:

- **OpenCV**
- **Optical Character Recognition (OCR)**

It detects:

- User interface elements
- Buttons
- Text fields
- Labels
- Menu components

This information is used to understand the current interface state.

Agentic AI Engine

The Agentic AI engine interprets **natural language instructions** and generates step-by-step execution plans.

Based on visual input and system state, the agent determines the next required action in order to complete the assigned task.

Execution Controller

The execution controller translates the agent's decision into **structured commands** that are sent to the ESP32 HID device.

These commands define specific actions such as mouse clicks, keyboard typing, or cursor movements.

6. Integration Layer Deployment

The **Integration Layer** connects the automation system with external platforms used for task management and monitoring.

Jira Integration

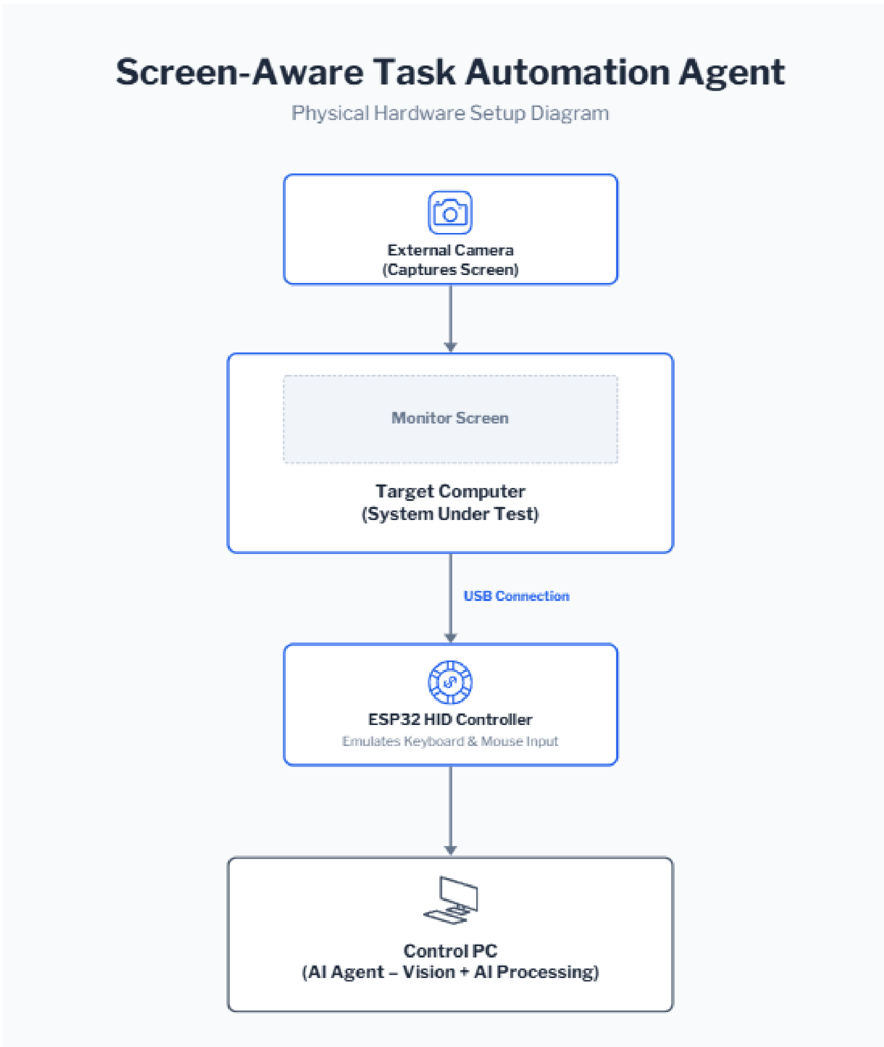
The system integrates with **Jira APIs** to record:

- Automation tasks
- Execution results
- Test outcomes
- Error logs

This integration enables automated test results to be tracked within a standard issue management workflow.

7. Physical Setup Diagram

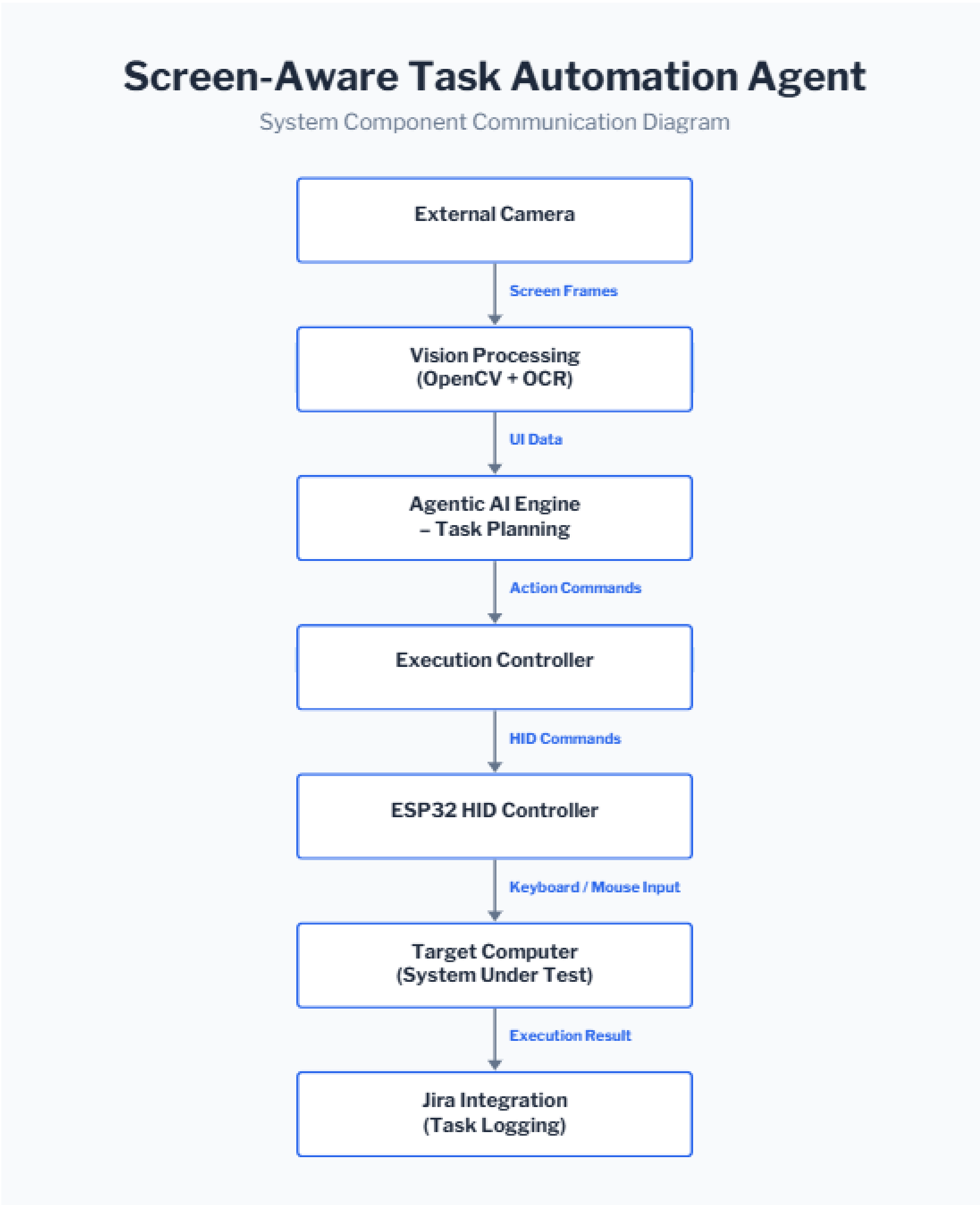
The physical setup of the system is illustrated below.



This setup ensures that the automation system can **observe and control the target computer externally** without installing software on the system under test.

8. Component Communication Diagram

The following diagram illustrates how system components communicate during task execution.



This communication flow ensures that each action performed by the system is **validated using visual feedback before proceeding to the next step**.

9. Operational Workflow

The automation system executes tasks through the following workflow:

1. The external camera captures the current state of the target computer screen.
2. The vision module analyzes the image and detects user interface components.
3. The Agentic AI engine interprets the task instructions and determines the next action.
4. The execution controller converts the action into a command format.
5. The ESP32 HID controller performs keyboard or mouse inputs on the target computer.
6. The system captures updated visual feedback to verify task completion.
7. The results of the automation task are logged through the integration layer.

10. Scalability Considerations

The proposed architecture supports future scalability and extension. Multiple automation environments can be deployed by connecting additional cameras and HID controllers.

Future improvements may include:

- Cloud-based AI processing
- Multiple target system orchestration
- Integration with **CI/CD pipelines**
- Advanced UI recognition techniques
- Automated regression testing frameworks

These enhancements would allow the system to scale for **large-scale automated testing environments**.